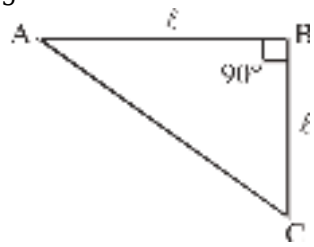
PHYSICS

1) A car moving along a straight road with speed of 144 km h^{-1} is brought to a stop within a distance of 200 m. How long does it take for the car to stop ?

(1) 5 s
(2) 10 s
(3) 15 s
(4) 20 s

2) Figure shows a thin metallic triangular sheet ABC. The mass of the sheet is M. The moment of inertia of the sheet about side AC is-



(1) $\frac{M\ell^2}{18}$
(2) $\frac{M\ell^2}{12}$
(3) $\frac{M\ell^2}{6}$
(4) $\frac{M\ell^2}{4}$

3) A sphere is released on a smooth inclined plane from the top. When it moves down its angular momentum is:

(1) conserved about every point
(2) conserved about the point of contact only
(3) conserved about the centre of the sphere only
(4) conserved about any point on a fixed line parallel to the inclined plane and passing through the centre of the ball.

4) A small stone is tied with a string which could break at tension 400 N. The stone is moving in horizontal plane with Angular speed ω = t. Then the time at which string could break is (Length of string is 1m and and mass of particle is 1kg)

(1) 30 sec.
(2) 25 sec.
(3) 20 sec.

(4) None

5) What should be the angular speed of the earth, so that a body lying on the equator may appear weightless ($g = 10m/s^2$, R = 6400 km) :-

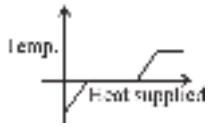
(1) $\frac{1}{800}$rad/s

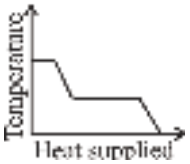
(2) $\frac{1}{400}$rad/s

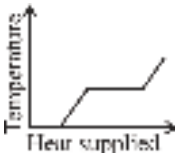
(3) $\frac{1}{600}$rad/s

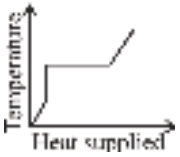
(4) $\frac{1}{100}$rad/s

6) A block of ice at –10ºC is slowly heated and converted to steam at 100ºC. Which of the following curves represents the phenomenon qualitatively :-

(1) 

(2) 

(3) 

(4) 

7) The radiation power emitted by a star A is 10000 times that of the sun. If the surface temperatures of the sun and star A are 6000 K and 2000 K respectively, the ratio of the radii of the star A and the sun is :-

(1) 300 : 1

(2) 600 : 1

(3) 900 : 1

(4) 1200 : 1

8)

A frictionless heat engine can be 100% efficient only if its exhaust temperature is :-

(1) 0°C

(2) 0K

(3) equal to input temperature

(4) half of its input temperature

9) A ball falls from height h. After 1 second, another ball falls freely from a point 20 m below the point from where the first ball falls. Both of them reach the ground at the same time. What is the value of h :-
($g = 10\ m/s^2$)

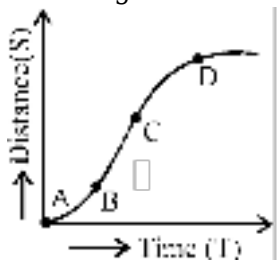
(1) 11.2 m

(2) 21.2 m

(3) 31.2 m

(4) 41.2 m

10) A particle shows distance-time curve as given in this figure. The maximum instantaneous velocity of the particle is around the point :-



(1) D

(2) A

(3) B

(4) C

11) A uniform chain of mass M and length L is lying on a frictionless table in such a way that its 1/3 part is hanging vertically down. The work done in pulling the chain up the table is

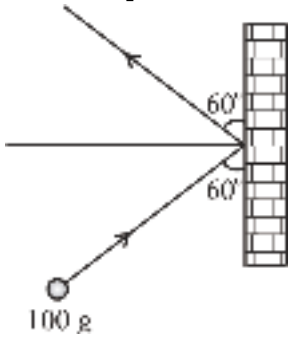
(1) $\frac{MgL}{3}$

(2) $\frac{MgL}{6}$

(3) $\frac{MgL}{9}$

(4) $\frac{MgL}{18}$

12) A mass of 100 g strikes the wall with speed 5 m/s at an angle as shown in figure and it rebounds with the same speed. If the contact time is 2×10^{-3} s, what is the force applied on the mass by the wall :-



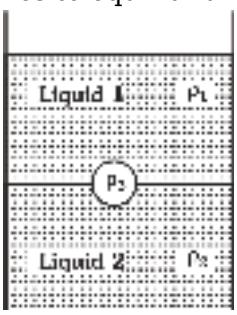
(1) $250\sqrt{3}$N to right

(2) 250 N to right

(3) $250\sqrt{3}$N to left

(4) 250 N to left

13) A jar is filled with two non-mixing liquids 1 and 2 having densities ρ_1 and ρ_2, respectively. A solid ball, made of a material of density ρ_3, is dropped in the jar. It comes to equilibrium in the position shown in the figure. Which of the following is true for ρ_1, ρ_2 & ρ_3



(1) $\rho_3 < \rho_1 < \rho_2$

(2) $\rho_1 > \rho_3 > \rho_2$

(3) $\rho_1 < \rho_2 < \rho_3$

(4) $\rho_1 < \rho_3 < \rho_2$

14) The radius of a soap bubble is increased from $\frac{1}{\sqrt{\pi}}$ cm to $\frac{2}{\sqrt{\pi}}$ cm. If the surface tension of water is 30 dynes per cm, then the work done will be :-

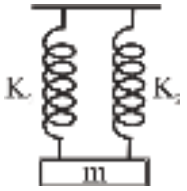
(1) 180 ergs

(2) 360 ergs

(3) 720 ergs

(4) 960 ergs

15) A mass is suspended separately by two different springs in successive order then time period is t_1 and t_2 respectively. If it is connected by both spring as shown in figure then time period is t_0, the correct relation is :–



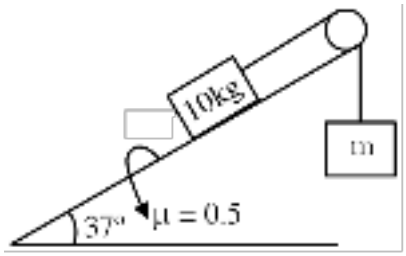
(1) $t_0^2 = t_1^2 + t_2^2$

(2) $t_0^{-2} = t_1^{-2} + t_2^{-2}$

(3) $t_0^{-1} = t_1^{-1} + t_2^{-1}$

(4) $t_0 = t_1 + t_2$

16) The maximum and minimum value of 'm' for which system remain in rest :



(1) 10 kg and 2 kg

(2) 6 kg and 4 kg

(3) 4 kg and 2 kg

(4) 6 kg and 2 kg

17) The stationary wave y = 2a sin kx cosωt in a stretched string is the result of superposition of y_1 = a sin(kx – ωt) and :-

(1) y_2 = a cos(kx + ωt)

(2) y_2 = a sin(kx + ωt)

(3) y_2 = a cos(kx – ωt)

(4) y_2 = a sin(kx – ωt)

18) If the ratio of amplitudes of two interfering waves in 4 : 3, then the ratio of maximum and minimum intensity is :–

(1) 16 : 18

(2) 18 : 16

(3) 49 : 1

(4) 94 : 1

19) A student measures the mass of an object five times. The data are 2.46 kg, 2.47 kg, 2.41 kg, 2.42 kg and 2.44 kg. The reported mean mass with error should be:

(1) 2.44 ± 0.01 kg

(2) 2.44 ± 0.02 kg

(3) 2.44 ± 0.03 kg

(4) 2.44 ± 0.04 kg

20) A soap film of surface tension 3×10^{-2} Nm^{-1} formed in rectangular frame, can support a straw. The length of the film is 10 cm. Mass of the straw the film can support is

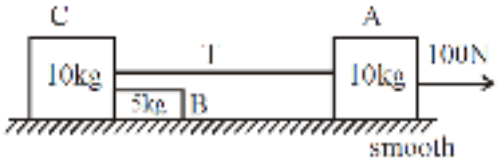
(1) 0.06 gm

(2) 0.6 gm

(3) 6 gm

(4) 60 gm

21) Contact force between block B & C is :-



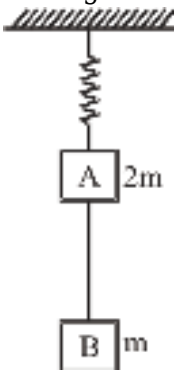
(1) 10 N

(2) 20 N

(3) 40 N

(4) 50 N

22) Two blocks A and B of masses 2m and m are suspended by means of massless spring and a massless inextensible string as shown in figure. The acceleration of blocks A and B in immediately after the string is cut respectively :-



(1) g downward ↓, g downward ↓

(2) g upward ↑, $\frac{g}{2}$ downward ↓

(3) $\frac{g}{2}$ downward ↓, 2g downward ↓

(4) $\frac{g}{2}$ upward ↑, g downward ↓

23) If at temperature T_1 = 1000 K, the wavelength corresponding to maximum radiation is 1.4×10^{-6} m, then at what temperature that wavelength will be 2.8×10^{-6} m

(1) 2000 K

(2) 500 K

(3) 250 K

(4) None of these

24) A physical quantity A is related to four observations a, b, c and d as follows, $A = \frac{a^2 b^3}{c\sqrt{d}}$. The percentage errors of measurement in a, b, c and d are 0.1%, 0.3%, 0.2% and 0.2% respectively. What is the percentage error in the quantity A

(1) 1.2%

(2) 0.7%

(3) 0.5%

(4) 1.4%

25)

For a diamagnetic material :-

(1) $\mu_r < 1,\ \square < 0$

(2) $\mu_r > 1,\ \square < 1$

(3) $\mu_r > 1,\ \square > 1$

(4) $\mu_r < 1,\ \square > 0$

26) An electron having kinetic energy K is moving in a circular path of radius R perpendicular to a uniform magnetic field. If kinetic energy is double and magnetic field tripled, the radius will become :-

(1) $\frac{2R}{3}$

(2) $\frac{\sqrt{2}}{3}R$

(3) $\sqrt{\frac{2}{3}}R$

(4) $\frac{2}{\sqrt{3}}R$

27) A convex mirror forms the image of an object such that the distance between the object and image is 60 cm and the magnification produced is 1/4. The focal length of the mirror will be :-

(1) 8.6 cm

(2) 6.2 cm

(3) 10 cm

(4) 16 cm

28) What is the respective number of α and β particles emitted in the following radioactive decay :-
$_{90}X^{200} \rightarrow {}_{80}Y^{168}$

(1) 6 and 8

(2) 8 and 8

(3) 6 and 6

(4) 8 and 6

29) **Statement-1 :** Wave length of microwaves lies between infrared and radiowaves.
Statement-2 : Microwave oven based on application of microwaves.

(1) Statement-1 is true, Statement-2 is true

(2) Statement-1 is false, Statement-2 is true

(3) Statement-1 is true, Statement-2 is false

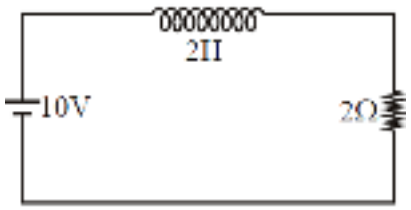
(4) Statement-1 is false, Statement-2 is false

30) A light wave is incident normally over a slit of width 24×10^{-5} cm. The angular position of second

dark fringe from the central maxima is 30°. What is wave length of light.

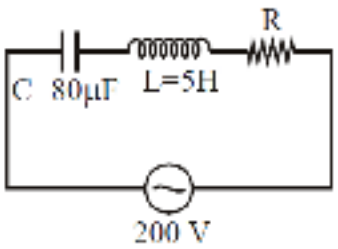
(1) 6000 Å
(2) 5000 Å
(3) 3000 Å
(4) 1500 Å

31) In the figure magnetic energy stored in the coil is:



(1) Zero
(2) Infinite
(3) 25 Joules
(4) None of the above

32) The source frequency for the given circuit at resonance is :



(1) 25 Hz
(2) $\frac{25}{\pi}$ Hz
(3) 50 Hz
(4) $\frac{50}{\pi}$ Hz

33) Calculate the angular dispersion produced by a thin prism of angle 12° if refractive index for red and violet colours are respectively 1.56 and 1.68.

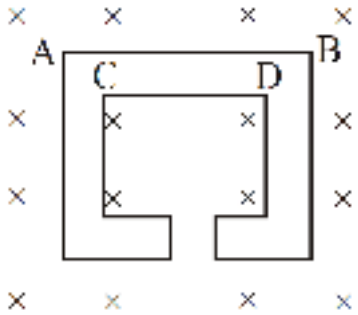
(1) 1.44°
(2) 2°
(3) 2.05°
(4) None of these

34) A resistor develops 600 J of heat in 20 seconds when a current of 3A is passed through it. When the current is increased to 4A, the amount of heat energy developed in 18 seconds will be :

(1) 280 J
(2) 360 J
(3) 720 J

(4) 960 J

35) A wire is bent to form the double loop shown in the figure. There is a uniform magnetic field directed into the plane of the loop. If the magnitude of this field is decreasing, current will flow from



(1) A to B and C to D

(2) B to A and D to C

(3) A to B and D to C

(4) B to A and C to D

36) Assume that an electric field $\vec{E} = 30x^2\hat{i}$ exists in space. Then the potential difference $V_A - V_O$, where V_O is the potential at the origin and V_A the potential at x = 2 m is :-

(1) – 80 V

(2) 80 V

(3) 120 V

(4) – 120 V

37) In a wire of cross-section radius r, free electrons travel with drift velocity v when a current flows through the wire. What is the current in another wire of half the radius and of the same material when the drift velocity is 2v ?

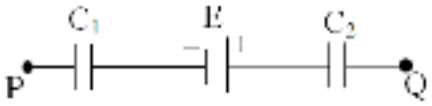
(1) 2I

(2) I

(3) I/2

(4) I/4

38) A circuit has a section PQ as shown in the figure with E = 20V, C_1 = 2μF, C_2 = 4μF and the



potential difference $V_P - V_Q$ = 10 V. The voltage across C_1 is :-

(1) Zero

(2) 10V

(3) 20 V

(4) 30 V

39) The maximum wavelength of radiation that can produce photoelectric effect in a certain metal is

200 nm. The maximum kinetic energy acquired by electron due to radiation of wavelength 100 nm will be :-

(1) 12.4 eV

(2) 6.2 eV

(3) 100 eV

(4) 200 eV

40) The de-Broglie wavelength of a neutron at 27°C is λ. What will be its wavelength at 927°C :-

(1) λ/2

(2) λ/3

(3) λ/4

(4) λ/9

41) n^{th} bright fringe of red light (λ_1 = 7500 Å) coincides with $(n + 1)^{th}$ bright fringe of green light (λ_2 = 6000 Å). The value of n = ?

(1) 2

(2) 3

(3) 4

(4) 5

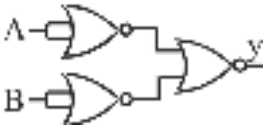
42) The nuclear radius of a nucleus with nucleon number 16 is 3×10^{-15} metre. Then the nuclear radius of a nucleus with nucleon number 128 is :-

(1) 3×10^{-15} m

(2) 1.5×10^{-15} m

(3) 6×10^{-15} m

(4) 4.5×10^{-15} m

43) Find the equivalent logic gate : 

(1) OR

(2) AND

(3) NOR

(4) NAND

44) Light travels in two media A and B with speed 1.8×10^8 ms^{-1} and 2.4×10^8 ms^{-1} respectively. Then the critical angle between them is :-

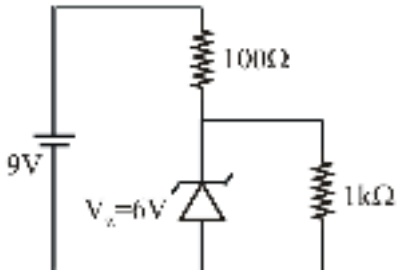
(1) $\sin^{-1}\left(\frac{2}{3}\right)$

(2) $\tan^{-1}\left(\frac{3}{4}\right)$

(3) $\tan^{-1}\left(\frac{2}{3}\right)$

(4) $\sin^{-1}\left(\frac{3}{4}\right)$

45) In the given circuit current through the zener diode is :-



(1) 30 mA

(2) 24 mA

(3) 36 mA

(4) 6 mA

CHEMISTRY

1) Nicotinic acid ($K_a = 1.4 \times 10^{-5}$) is represented by formula HNiC. Its percentage dissociation in a solution containing 0.1 mole nicotinic acid in 2L solution will be :-

(1) 2.82%

(2) 3.34%

(3) 1.67%

(4) 2.14%

2) Which of the following reaction represent heat of formation :-

(1) $2C(\text{graphite, s}) + O_2(g) \rightarrow 2CO(g)$

(2) $C(\text{diamond, s}) + 2H_2(g) \rightarrow CH_4(g)$

(3) $C(\text{graphite, s}) + 2H_2(g) \rightarrow CH_4(g)$

(4) $H^+(aq) + OH^-(aq) \rightarrow H_2O(\square)$

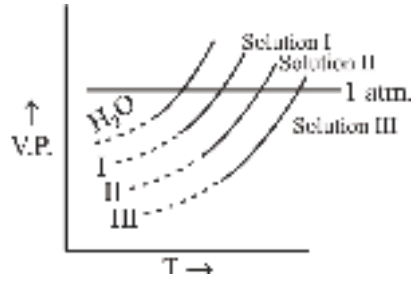
3) The electric charge for electrode deposition of 1g equivalent of a substance is :-

(1) 1 ampere per second

(2) 96500 coulombs per second

(3) 1 ampere for one hour

(4) The charge on one mol of electrons

4) 

Which solution have highest elevation in boiling point :

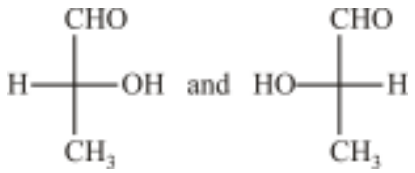
(1) H_2O

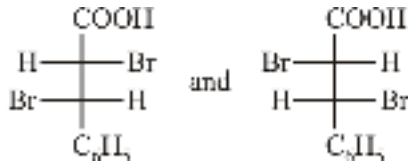
(2) Solution I

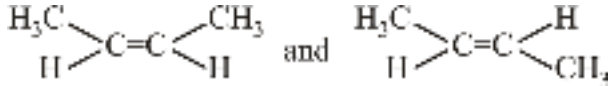
(3) Solution II

(4) Solution III

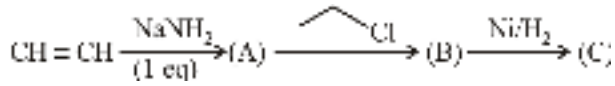
5) Which of the following pair is diastereomers:-

(1) 

(2) 

(3) 

(4) All of these

6) 

Product C is

(1) 

(2) 

(3) 

(4) 

7)

In Ziese's salt, (C=C) bond length is :-
Note : C–C bond length in ethane is 1.54Å
C=C bond length in ethene is 1.34Å

C≡C bond length in ethyne is 1.20Å

(1) 1.37 Å

(2) 1.19 Å

(3) 1.87Å

(4) 1.34Å

8) Oxidation number of sulphur in caro's acid (H_2SO_5) is :-

(1) +5
(2) +6
(3) +7
(4) +8

9) Which relation of electromotive force (emf) of an electrochemical cell is correct :-

(1) emf of cell = oxidation potential of anode – reduction potential of cathode
(2) emf of cell = oxidation potential of anode + reduction potential of cathode
(3) emf of cell = reduction potential of anode + reduction potential of cathode
(4) All of these

10) 1-Phenyl-2-chloropropane when reacted with alc. KOH mainly gives :-

(1) 1-phenyl propene
(2) 3-phenyl propene
(3) 1-phenyl-2-propanol
(4) 3-phenyl-2-propene

11) The bond energy order of He_2^+ and HeH^+ is :-

(1) $He_2^+ > HeH^+$
(2) $HeH^+ > He_2^+$
(3) $HeH^+ = He_2^+$
(4) Can not be predicted

12) Which of the following option is not incorrect ?

(1) Stability in aqueous medium $Cr^{3+} < Mn^{3+}$
(2) $Fe + HCl \rightarrow FeCl_2 + H_2$
(3) $CrO_4^{2-} \xrightarrow{OH^{\ominus}} Cr_2O_7^{2-}$
(4) Alloy brass and bronze both contain Sn.

13) Electronic configuration of valence shell of four elements A, B, C and D are :-
(A) $ns^2 np^1$ (B) $ns^2 np^2$
(C) $ns^2 np^3$ (D) $ns^2 np^4$
Correct increasing order of their second ionisation energies is :-

(1) A < B < C < D
(2) A < B < D < C
(3) B < A < D < C
(4) B < A < C < D

14)

Select the compounds/species those can be exists :

H_2S, OF_6, BF_4^-, PH_5, XeF_5, CCl_6^{-2}, PCl_5

(1) 3

(2) 4

(3) 5

(4) 2

15) Match List I with List II.

List-I (Complex)		**List-II (Type of isomerism)**	
A.	$[Co(NH_3)_5(NO_2)]Cl_2$	I.	Solvate isomerism
B.	$[Co(NH_3)_5(SO_4)]Br$	II.	Linkage isomerism
C.	$[Co(NH_3)_6][Cr(CN)_6]$	III.	Ionization isomerism
D.	$[Co(H_2O)_6]Cl_3$	IV.	Coordination isomerism

Choose the **correct** answer from the options given below :

(1) A-II, B-III, C-IV, D-I

(2) A-I, B-III, C-IV, D-II

(3) A-I, B-IV, C-III, D-II

(4) A-II, B-IV, C-III, D-I

16) **Assertion :** $KMnO_4$ is purple in colour due to charge transfer.
Reason : There is no electron present in d-orbitals of Manganese in MnO_4^-.

(1) Both **Assertion** and **Reason** are true but **Reason** is NOT a correct explanation of **Assertion**.

(2) **Assertion** is true but **Reason** is false.

(3) **Assertion** is false but **Reason** is true.

(4) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.

17) **Assertion :** PbI_4, $BiCl_5$, FeI_3, CuI_2 are not exist.
Reason : Higher oxidation state of metal ions are strong oxidising agent which give redox reaction with Iodide ion.

(1) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.

(2) **Assertion** is true but **Reason** is false.

(3) **Assertion** is false but **Reason** is true.

(4) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.

18) Which of the following statements is correct for glucose :-

(1) It gives positive test with schiff's reagent

(2) It react's with $NaHSO_3$ and NH_3

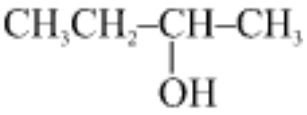
(3) It gives Positive test with 2, 4-DNP

(4) It gives positive test with fehling solution

19) In following given reaction

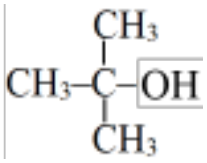
$CH_3COOH + PCl_5 \rightarrow (A) \xrightarrow[(2)\ H_2O/H^+]{(1)\ CH_3MgBr(excess)} (B)$

Product B would be :-

(1) 

(2) $CH_3COC_6H_5$

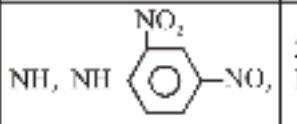
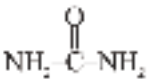
(3) $CH_3CH(OH)CH_3$

(4) 

20)

Which of the following product is not correct :

Carbonyl compound + Reagent → Product

S.N.	Reagent	Product
(1)	NH_3	Imine
(2)	NH_2OH	Oxime
(3)		2,4-Dinitrophenyl hydrazone
(4)		Semicarbazone

(1) 1

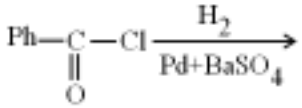
(2) 2

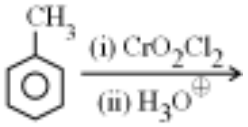
(3) 3

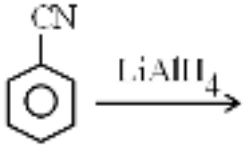
(4) 4

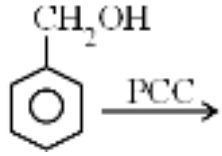
21)

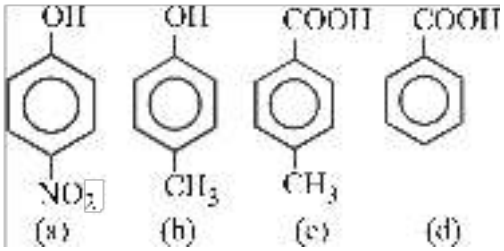
Which of the following reaction can not be used for preparation of benzaldehyde ?

(1) 

(2) 

(3) 

(4) 

22) 

Compare acidic strength

(1) a > c > d > b

(2) d > a > c > b

(3) d > c > a > b

(4) d > c > b > a

23) Incorrect reaction is -

(1) $CH_4 + O_2 \xrightarrow[523/100atm]{Cu} CH_3OH$

(2) $CH_4 + O_2 \xrightarrow[\Delta]{Mo_2O_3} HCHO$

(3) $CH_3CH_3 + O_2 \xrightarrow[\Delta]{Mn(OAc)_2} CH_3COOH$

(4) $(CH_3)_2CH_2 \xrightarrow[Oxidation]{KMnO_4} (CH_3)_2CH-OH$

24) Consider following table–

Column-I

(A) Best and latest technique for isolation, purification and separation of organic compounds

(B) In Lassaigne's test for 'N'

(C) In Lassaigne's test for 'S'

(D) Aniline is purified from water by

(E) Glycerol is purified from spent lye

Column-II

(i) Chromatography

(ii) Vacuum distillation

(iii) Violet coloured complex is formed

(iv) Prussian blue coloured complex is formed

(v) Steam distillation

The correct match is–

	(A)	(B)	(C)	(D)	(E)
(1)	(i)	(ii)	(iii)	(iv)	(v)
(2)	(iv)	(v)	(iii)	(ii)	(i)
(3)	(i)	(iv)	(iii)	(v)	(ii)
(4)	(v)	(iii)	(iv)	(ii)	(i)

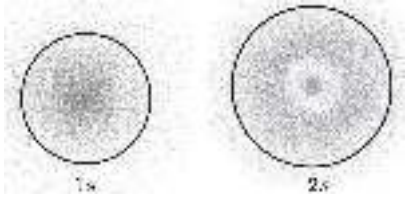
(1) 1

(2) 2

(3) 3

(4) 4

25) The probability density plots of 1s and 2s orbitals are given in figure:



The density of dots in a region represents the probability density of finding electrons in the region. On the basis of above diagram which of the following statements is/are correct?

(i) 1s and 2s orbitals are spherical in shape.

(ii) The probability density of finding the electron is minimum near the nucleus.

(iii) The probability of finding the electron at a given distance is equal in all directions.

(iv) The probability density of electrons for 2s-orbital decreases uniformly as distance from the nucleus increases.

(1) (i) & (ii) only

(2) (ii) & (iii) only

(3) (i) & (iii) only

(4) All are correct

26) If $CO + H_2O \rightleftharpoons CO_2 + H_2$, K

$2\ CO + O_2 \rightleftharpoons 2CO_2$, K_1

$2\ H_2 + O_2 \rightleftharpoons 2H_2O$, K_2

Then correct relation will be :-

(1) $K = \sqrt{\frac{K_1}{K_2}}$

(2) $K = \sqrt{K_1 \times K_2}$

(3) $K = K_1 \times K_2$

(4) None

27) Match the columns

	Column-I		**Column-II**

(A)	Number of collisions per second per unit volume of the reaction mixture.	(p)	Effective collisions.
(B)	Fraction of molecules with energies equal to or greater than E_a	(q)	Collision frequency
(C)	Molecules for which $Rate = Z_{AB}e^{-E_a/RT}$ shows significant deviations	(r)	$e^{-E_a/RT}$
(D)	Collision in which molecules collide with sufficient K.E. and proper orientation.	(s)	Complex molecules

(1) A-(q),B-(r),C-(s),D-(p)

(2) A-(r),B-(q),C-(s),D-(p)

(3) A-(q),B-(s),C-(r),D-(p)

(4) A-(q),B-(r),C-(p),D-(s)

28) Match list-I with list-II

	List-I		**List-II**
(A)	68% nitric acid and 32% water by mass	(I)	Minimum boiling azeotrope
(B)	Solution showing large positive deviation from Raoult's law may form	(II)	Ideal solution
(C)	n-hexane and n-heptane	(III)	Maximum boiling azeotrope
(D)	Two solutions having same osmotic pressure at a given temperature	(IV)	Isotonic solution

Choose the correct answer from the options given below :-

(1) (A)-I, (B)-III, (C)-II, (D)-IV

(2) (A)-III, (B)-IV, (C)-II, (D)-I

(3) (A)-III, (B)-I, (C)-IV, (D)-II

(4) (A)-III, (B)-I, (C)-II, (D)-IV

29) On the basis of following data, select the correct pH value for 10^{-2} M NH_4OH(aq) solution at 25°C

$\Lambda_m(NH_4OH)$ of 10^{-2} M solution = 10.2 S-cm^2 mol^{-1}

$\Lambda^o_m(NH_4Cl)$ = 130 S-cm^2 mol^{-1}

$\Lambda^o_m(NaOH)$ = 320 S-cm^2 mol^{-1}

$\Lambda^o_m(NaCl)$ = 110 S-cm^2 mol^{-1}

(1) 3.52

(2) 10.48

(3) 2

(4) 12

30) Match List-I with List-II

	List-I (Compounds/ions)		List-II (Oxidation state of underlined atom)
(A)	$[\underline{C}r(NH_3)_4Cl_2]^+$	(I)	–3
(B)	$\underline{N}H_4Cl$	(II)	+6
(C)	$H_2\underline{S}_2O_8$	(III)	+3
(D)	$\underline{Os}O_4$	(IV)	+8

Choose the correct answer from the options given below :

(1) A-III, B-I, C-IV, D-II

(2) A-III, B-I, C-II, D-IV

(3) A-I, B-II, C-III, D-IV

(4) A-II, B-I, C-IV, D-III

31) Match the column I with column II and select correct answer.

Column I (symbol of element)		Column II (atomic number of element)	
(A)	Hs	(1)	118
(B)	Rg	(2)	108
(C)	Mc	(3)	111
(D)	Og	(4)	115

(1) A-2, B-1,C-4,D-3

(2) A-2, B-3,C-4,D-1

(3) A-3, B-2,C-1,D-1

(4) A-1, B-3,C-4,D-1

32) Which of the following set of molecules have zero dipole moment :

(1) CO_2, HBr, BF_3, H_2

(2) CF_4, BF_3, CO_2, PCl_5

(3) H_2, HI, $CHCl_3$, CH_4

(4) CH_4, CO_2, Cl_2, HI

33) **Assertion :** Transition metals and their compounds are good catalyst
Reason : They have the ability to adopt multiple oxidation state.

(1) Both Assertion and Reason are true and the Reason is the correct explanation of Assertion

(2) Both Assertion and Reason are true and the Reason is not the correct explanation of Assertion

(3) Assertion is true but Reason is false

(4) Both Assertion and Reason are false

34) Which one have maximum third ionisation enthalpy :

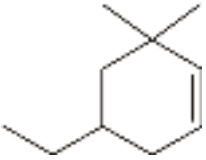
(1) Ti
(2) V
(3) Cr
(4) Mn

35) The spin only magnetic moment of $[MnBr_4]^{-2}$ in Bohr magneton :

(1) 5.84
(2) 4.84
(3) 2.85
(4) 1.73

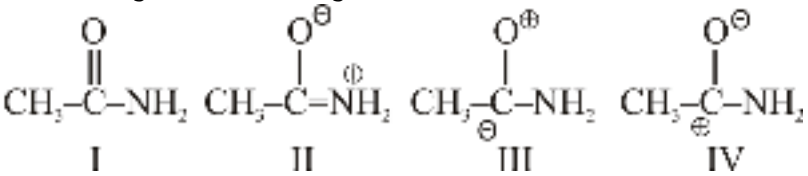
36) **Statement-I :** When H_2S gas is passed through sodium-nitroprusside solution it gives purple colour.
Statement-II : When K_2CrO_4 solution treated with barium acetate it gives red precipitate.

(1) Statement-I is false but statement-II is true
(2) Both statement-I and II are false
(3) Statement-I is true but statement-II is false
(4) Both statement-I and II are true

37) The IUPAC name of compound is 

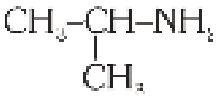
(1) 5-Ethyl-3,3-dimethyl cyclohexene
(2) 4-Ethyl-6,6-dimethyl cyclohex -2-ene
(3) 3-Ethyl-6,6-dimethyl cyclohex-6-ene
(4) 6-Ethyl-4,4-dimethyl cyclohexene

38) Arrange the following canonical structure in order of their stability :-

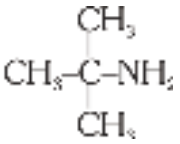


(1) IV > III > II > I
(2) I > II > III > IV
(3) I > II > IV > III
(4) III > II > I > IV

39) Which compound does not respond to carbylamine reaction :-

(1) 

(2) C_2H_5–NH–C_2H_5

(3) 

(4) Ph–NH_2

40) In Duma's method for estimation of nitrogen, 0.25 g of an organic compound gave 40 mL of nitrogen collected at 300 K temperature and 725 mm pressure. If the aqueous tension at 300 K is 25 mm, the percentage of nitrogen in the compound is :-

(1) 18.20

(2) 16.76

(3) 15.76

(4) 17.36

41) In the Williamson synthesis of ethers given by the general equation :
R–X + R–ONa → R–O–R. The yield from R–X follows the sequence :-

(1) $CH_3 > 1° > 2° > 3°$

(2) $CH_3 < 1° < 2° < 3°$

(3) $CH_3 < 1° < 2° > 3°$

(4) $CH_3 > 1° < 2° < 3°$

42) Solubility of AgCl(s) with solubility product 1.6×10^{-10} in 0.1 M $MgCl_2$ solution would be :

(1) 1.6×10^{-9} M

(2) 8×10^{-9} M

(3) 1.26×10^{-5} M

(4) 8×10^{-10} M

43) For a reaction X → Y, rate constant is 1.5 M min^{-1}, if 75% of reaction is completed in 50 minutes, then what % of reactant will be left unreacted after 100 minutes :

(1) 6.25

(2) 12.5

(3) 0

(4) 18.75

44) On dissolving 0.5 g of a non-volatile, non-ionic solute in 39 g benzene, its vapour pressure decreases from 650 mmHg to 640 mmHg, then depression in freezing point of benzene (in K) upon addition of solute is :
(Given K_f(benzene) = 5.12 K-kg mol^{-1})

(1) 0.51

(2) 1.02

(3) 5.12

(4) 2.56

45) Select the correct statements from the following :
(A) In titration of $KMnO_4$ and oxalic acid, $KMnO_4$ act as self indicator
(B) In half reaction,
$MnO_4^- + xH^+ + ye^- \rightarrow Mn^{+2} + zH_2O$
x, y and z are respectively 5, 8 and 4.
(C) A solution of known concentration is called a standard solution.
(D) A lyophilic sol is more stable than lyophobic sol.
(E) Reagent of known concentration which is taken in burette is titer.
Choose the correct answer from the options given below :

(1) C, D and E only

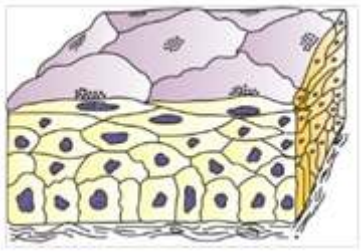
(2) A and D only

(3) A, C and D only

(4) A, B, C and D only

BIOLOGY

1)



Which of the following statements is true about above given diagram :-

(1) It is composed of tall and single layered cells

(2) It has limited role in secretion and absorption

(3) It is found in nephron

(4) Their main function is to form diffusion boundaries

2) Testis of cockroach located in -

(1) I^{st}-3rd abdominal segments

(2) 2nd-5th abdominal segments

(3) 4th-6th abdominal segments

(4) 2nd-4th abdominal segments

3) When frog is on land then respiration takes place by ?

(1) Only skin

(2) Skin and buccal cavity

(3) Skin, buccal cavity and lungs

(4) Gill

4) Which of the following is/are similarity between *Periplaneta* and *Pheretima* ?
(A) Open blood vascular system
(B) Internal fertilisation
(C) Dioecious
(D) Double ventral nerve cord

(1) Only B, C

(2) B, C and D

(3) Only D

(4) A, C and D

5) Which of the following are incorrect for platyhelminthes ?

(1) All are endoparasite

(2) They are bilateral & Triploblastic

(3) Hooks and suckers are present in the parasitic forms

(4) They have dorso-ventrally flattened body.

6) Between carpal and metacarpal of thumb____joint present :-

(1) Ball and socket

(2) Hinge

(3) Pivot

(4) Saddle

7) Vasoconstriction causes :-

(1) Increase in heart beat

(2) Decrease in heart beat

(3) Increase in blood pressure

(4) Decrease in blood pressure

8) Read the following four statements (A-D) :-
(A) In amphibia and reptiles incomplete double circulation present.
(B) In arthropods and molluscs open circulatory system is present.
(C) In joint diastole stage semilunar valves are closed.
(D) The hepatic portal vein carries blood from intestine to the kidney.
How many of the above statements are wrong?

(1) Four

(2) One

(3) Two

(4) Three

9) Functional residual capacity (FRC) is :-

(1) TV + ERV
(2) TV + ERV + IRV
(3) ERV + RV
(4) TV + ERV + IRV + RV

10) Which of the following is present in pons ?

(1) Dorsal respiratory centre
(2) Ventral respiratory centre
(3) Pneumotaxic centre
(4) Chemosensitive centre

11) **Statement-I :** Synaptic cleft is always present in chemical synapses.
Statement-II : The electrical potential difference across the resting plasma membrane is called as resting potential.

(1) Statement-I is correct, statement-II is incorrect.
(2) Statement-II is correct, statement-I is incorrect.
(3) Both statement-I and II are correct.
(4) Both statement-I and II are incorrect.

12) When a neuron is in resting state i.e. not conducting any impulse, the axonal membrane is :-

(1) Comparatively more permeable to K^+ ions and nearly impermeable to Na^+ ions.
(2) Comparatively more pemeable to Na^+ ions and nearly impermeable to K^+ ions.
(3) Equally permeable to both Na^+ and K^+ ions.
(4) Impermeable to both Na^+ and K^+ ions.

13) ***Assertion :-*** Analysis of urine helps in clinical diagnosis of many metabolic disorders.
Reason :- Glycosuria is an indicative of diabetes insipidus.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

14)

Which one of the following options gives the correct categorisation of six animals according to the type of nitrogenous wastes (A, B, C), they give out

	A	**B**	**C**
	AMMONOTELIC	**UREOTELIC**	**URICOTELIC**

(1)	Aquatic amphibia	Frog, Humans	Pigeon, Lizards, Cockroach
(2)	Aquatic amphibia	Cockroach, Humans	Frog, Pigeon, Lizards
(3)	Pigeon, humans	Aquatic Amphibia, Lizards	Cockroach, Frog
(4)	Frog, Lizards	Aquatic Amphibia, Humans	Cockroach, Pigeon

(1) 1

(2) 2

(3) 3

(4) 4

15) Ball and socket joints can be seen in :-

(1) Wrist

(2) Fingers

(3) Neck

(4) Shoulders

16) Which one of the following is a flat bone on the ventral midline of thorax ?

(1) Scapula

(2) Sternum

(3) Femur

(4) Patella

17)

Match the column I with column II.

Column-I		**Column-II**	
A	Growth hormone	i	Ovulation
B	Luteinizing hormone	ii	Stimulate sertoli cells
C	Prolactin hormone	iii	Diabetogenic hormone
D	FSH	iv	Milk formation

(1) A → iii, B → ii, C → iv, D → i

(2) A → iii, B → i, C → iv, D → ii

(3) A → iii, B → ii, C → i, D → iv

(4) A → iii, B → iv, C → i, D → ii

18) Hormone which acts on the melanocytes (melanin containing cells) and regulates pigmentation of the skin is :-

(1) Melanocyte stimulating hormone

(2) ACTH

(3) Growth hormone

(4) Prolactin

19) Which of the following hormone is responsible for ovulation :-

(1) Progesterone

(2) Oestrogen

(3) Oxytocin

(4) Relaxin

20) Early Greek thinkers thought units of life called spores were transferred to different planets including earth. This statement belongs to which theory ?

(1) Oparin Haldane theory

(2) Bigbang theory

(3) Panspermia theory

(4) Abiogensis

21) Fore limbs of whale, bat, cheetah and humans share similarities in pattern of bones of fore limbs. Hence in these animals same structure developed along different directions due to adaptation to different needs. This evolution is called ?

(1) Divergent evolution

(2) Convergent evolution

(3) Co-evolution

(4) Natural selection

22) Which hormone stimulates milk ejection from the mammary gland in females ?

(1) Oxytocin

(2) Vasopressin

(3) Prolactin

(4) Both (1) and (3)

23) The functions of male sex accessory ducts and glands are maintained by :

(1) FSH

(2) LH

(3) Androgen

(4) ICSH

24) Which of the following statements incorrect about oogenesis ?

(1) No more oogonia are formed and added after birth

(2) It is initiated during the embryonic development of female

(3) First meiotic division get temporarily arrested at prophase-I stage

(4) It is results in the formation of four haploid gametes

25) During pregnancy most of the major organ systems are formed :-

(1) By the end of 12 weeks of pregnancy

(2) By the end of second month of pregnancy

(3) By the end of about 24 weeks of pregnancy

(4) By the end of nine months of pregnancy

26) **Assertion (A) :–** Antivenom and antiserum are examples of passive immunisation.
Reason (R) :– Anti-venom which is given to patients contains pre formed antibodies against snake venom.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

27) Match the following and choose the correct answer :

(A)	Fertilization	(i)	Fundus of uterus
(B)	Implantation	(ii)	Ovary
(C)	Functional maturation of sperms	(iii)	Ampulla
(D)	Oogenesis	(iv)	Epididymis

(1) A-(ii), B-(iii), C-(i), D-(iv)

(2) A-(ii), B-(i), C-(iv), D-(iii)

(3) A-(iii), B-(ii), C-(iv), D-(i)

(4) A-(iii), B-(i), C-(iv), D-(ii)

28) Given below are two statements –
Statement-I : Malignant tumors exhibit metastasis.
Statement-II : Contact inhibition is the property of malignant tumors.

(1) Both Statement-I and Statement-II are incorrect

(2) Statement-I is correct but Statement-II is incorrect

(3) Statement-I is incorrect but Statement-II is correct

(4) Both Statement-I and Statement-II are correct

29) Which of the following hormones are secreted in women only during pregnancy ?

(a) Relaxin (b) Oxytocin
(c) hCG (d) hPL
(e) estrogen

(1) (b) and (e) only
(2) (b), (c) and (e) only
(3) (a), (c) and (d) only
(4) (a), (b) and (e) only

30) Which of the following is an example of innate immunity ?

(1) B-lymphocyte
(2) T-lymphocyte
(3) Natural killer cell
(4) Antibody

31) Match the columns-I, II and III and choose the correct combination from the given option :-

Column-I		**Column-II**		**Column-III**	
(a)	Primary producer	(1)	T_1	K	Grass hopper
(b)	Primary consumer	(2)	T_2	L	Wolf
(c)	Secondary consumer	(3)	T_3	M	Man
(d)	Tertiary consumer	(4)	T_4	N	Grass

(1) a-1-K, b-2-L, c-3-M, d-4-N
(2) a-1-N, b-2-M, c-3-K, d-4-N
(3) a-1-N, b-2-K, c-3-L, d-4-M
(4) a-2-N, b-3-K, c-4-L, d-1-N

32) Which is not true regarding ecosystem ?

(1) Cycling of energy occurs
(2) Input of energy is required
(3) Self sustainable
(4) Cyclic exchange of materials between living beings and environment

33) **Assertion :** Mycorrhizae are considered mutualistic relationship between fungus and roots of higher plants.
Reason : This interactions confers benefit on both interacting species, where fungus helps the plant in absorption of essential nutrients while plant provides food to fungus.

(1) Both assertion & reason are true & the reason is a correct explanation of the assertion.
(2) Both assertion & reason are true but reason is not a correct explanation of the assertion.
(3) Assertion is true but the reason is false.
(4) Both assertion & reason are false.

34) Which of the following is not an example of hotspot in India?

(1) Western ghat
(2) Eastern ghat
(3) Himalaya
(4) Indo-Burma

35) How many varieties of mango has been estimated to be present in India ?

(1) 20000
(2) 50000
(3) 1000
(4) 2000000

36) Read the following statements :
(a) Predators keep prey population under control.
(b) Nearly 25 precent of all insects are known to be phytophagous.
(c) Predation, parasitism and commensalism share a common characteristic the interacting species do not live closely together.
(d) Endoparasites live inside the host body
How many statements are correct ?

(1) 1
(2) 2
(3) 3
(4) 4

37) Who explained that cells divided and new cells are formed from pre-existing cells :

(1) Schleiden
(2) Schwann
(3) R. Virchow
(4) R. Brown

38) Subunits of 80 s ribosome are :-

(1) 60 S + 40 S
(2) 50 S + 30 S
(3) 60 S + 20 S
(4) 40 S + 40 S

39) DNA replication and centriole duplication occurs in which phase of cell cycle :-

(1) G_1 phase
(2) G_2 phase
(3) S phase

(4) G_0 phase

40) Which of the following is true regarding secondary constriction present in chromosomes?

(1) This is darkly stained part of chromosome
(2) This is present on few chromosomes
(3) Position of it is highly variable
(4) Sometime kinetochore is present on it

41) Which of the following is not a significance of meiosis ?

(1) It maintains constant number of chromosomes in different cells of an organism
(2) It leads to genetic variability in the population of sexually reproducing organism
(3) It conserves the specific chromosome number of each species across generations in sexually reproducing organism
(4) It is involved in process of evolution

42) The rate of formation of new organic matter by consumer is called as :-

(1) Gross primary productivity
(2) Net primary productivity
(3) Secondary productivity
(4) Primary productivity

43) Choose the correct equation for 'species-area relationship' :-

(1) $\log A = \log C + Z \log S$
(2) $\log C = \log S + Z \log A$
(3) $\log Z = A \log C + \log S$
(4) $\log S = \log C + Z \log A$

44) Read the following statements and identify the correct answer from the options given below -
(i) All cells arise from pre-existing cells.
(ii) Ribosomes are membrane bound organelles.
(iii) The largest isolated single cell is the egg of an ostrich.
(iv) All eukaryotic cells are not identical

(1) Only (i) and (ii) are correct.
(2) Only (ii), (iii) and (iv) are correct.
(3) All are correct
(4) All are correct except (ii)

45) Which of the following floral feature is shown by both floral formula and floral diagram ?

(1) Aestivation of sepals in floral bud
(2) Adhesion between parts of floral whorls
(3) Number of locule/s in ovary

(4) Arrangement of ovule(s) within the ovary

46) In which of the following, ovary is inferior?

(1) Chinarose
(2) Mustard
(3) Guava
(4) Rose

47) Which of the following is not an event of sexual reproduction ?

(1) Transfer of gametes
(2) Fusion of gametes
(3) Formation of new organism from unfertilized female gamete
(4) Development of embryo from zygote

48) Select incorrect statement ?

(1) In mature pollen grain, the vegetative cell is bigger, has abundant food reserve and a large irregularly shaped nucleus
(2) Multicarpellary apocarpous gynoecium is found in *Michelia*
(3) Seeds are produced without fertilization in apomixis
(4) Seeds of black pepper are not perispermic

49) (A) In C_3-plants, if O_2 does bind to RuBisCO then CO_2 fixation decreases.
(B) C_4-plants have a mechanism that increases the concentration of O_2 at the RuBisCO enzyme site.
(C) In C_4-plants, RuBisCO functions as a oxygenase enzyme and minimising the carboxylase activity.
(D) C_4-plants lack photoxidation of water
(E) RuBisCO has a much greater affinity for CO_2.
Choose the option which have all the correct statements (from A to E) for photosynthesis :-

(1) A, B and C
(2) A, C and E
(3) B, D and E
(4) A and E

50) In which kingdom boundaries are not well defined ?

(1) Monera
(2) Protista
(3) Fungi
(4) Plantae

51) Which of the following is heterosporus pteridophyte ?

(1) *Selaginella*
(2) *Cycas*

(3) *Pinus*

(4) *Cedrus*

52) Which statement is incorrect ?

(1) In brown algae sexual reproduction may be isogamous, anisogamous and oogamous

(2) The liverworts grow usually in moist, shady habitats

(3) In gymnosperms ovules are enclosed by ovary wall

(4) In pteridophytes main plant body is sporophyte

53) Read the following statements carefully :-
(A) Epiblast is found in grass embryo.
(B) In typical angiospermic fertilized embryosac primary endosperm cell is triploid.
(C) Seeds of pea and groundnut are albuminous.
(D) Apple and strawberry are example of false fruits.
(E) A recent record of 2000 years old viable seed is of the *Phoenix dactylifera*.
How many above statements are correct :-

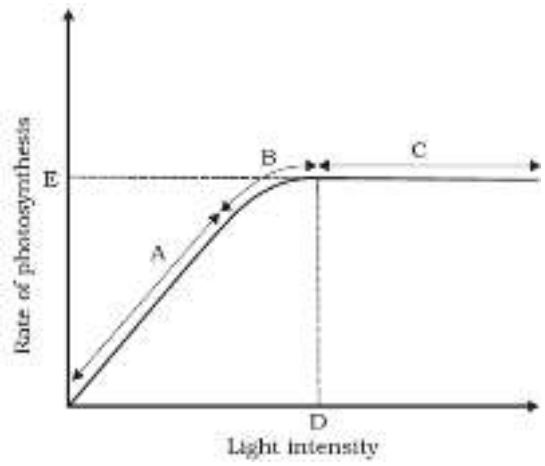
(1) Three

(2) Two

(3) Four

(4) Five

54)



Choose the incorrect option for given figure :-

(1) E-Light intensity, where maximum photosynthesis occurs

(2) C-factors other than light are limiting

(3) A-Light is limiting

(4) B-Transition between optimum and limiting value of light

55) Choose the correct option for krebs cycle for substrate and respective events in aerobic respiration :-

(1) Glucose, 4 oxidation, 3 CO_2 release

(2) Pyruvic acid, 4 oxidation, 3 CO_2 release

(3) Pyruvic acid , 3 oxidation, 2 CO_2 release

(4) Acetyl Co-A, 4 oxidation, 2 CO_2 release

56) Which growth hormone is responsible for initiating rooting in stem cutting :-

(1) Auxins
(2) Gibberellins
(3) Cytokinins
(4) Ethylene

57) Identify the incorrect option :-

	Class	**Fungi**	**Informations**
(1)	Deuteromycetes	*Alternaria*	Imperfect fungi
(2)	Basidiomycetes	*Agaricus*	Mushroom
(3)	Phycomycetes	*Albugo*	Smut
(4)	Ascomycetes	*Penicillium*	Antibiotic

(1) 1
(2) 2
(3) 3
(4) 4

58) In which of the following ploidy is correct :-
(A) Protonemal cells of moss (n)
(B) Prothallus of fern (2n)
(C) Gemma cells (n)
(D) Ovum of liverwort (2n)
(E) Leaves of moss (n)
(F) Zygote of fern (2n)

(1) A, C, E and F
(2) B, D, E and F
(3) A, B, D and F
(4) C, D, E and F

59) Which of the following is found among biomacromolecules fraction but it is not strictly a biomacromolecule :-

(1) Protein
(2) Lipid
(3) Polysaccharide
(4) Nucleic acid

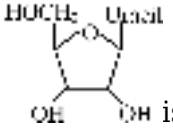
60) Which of the following secondary metabolite pair represents "polymeric substances":-

(1) Ricin and Abrin

(2) Morphine, Codeine

(3) Rubber, Gum

(4) Rubber, Glucose

61) Read the following statements carefully and find out pair of statements which are not correct :-
(a) Some secondary metabolites have ecological importance
(b) NaCl, $CaCO_3$, PO_4^-, CH_3COOH, CH_4 etc are inorganic constituents of living cell
(c)  is a nucleoside as well as found in RNA along with phosphate
(d) Secondary structure of proteins is absolutely necessary for the many biological activities of proteins

(1) a, b

(2) b, c

(3) c, d

(4) d, b

62) Match Column-I and Column-II.

	Column-I		**Column-II**
A.	Adenine derivative.	I.	ABA.
B.	Gaseous nature.	II.	Cytokinin.
C.	Terpenes.	III.	Auxin.
D.	Indole compounds.	IV.	Gibberellin.
E.	Carotenoid derivative.	V.	Ethylene.

(1) A-IV, B-V, C-I, D-III, E-II

(2) A-II, B-V, C-III, D-I, E-IV.

(3) A-III, B-V, C-IV, D-II, E-I.

(4) A-II, B-V, C-IV, D-III, E-I

63) Which of the following cofactors of enzymes are organic or inorganic compounds and are tightly bound to the apoenzyme?

(1) Metal ion

(2) Coenzyme

(3) Prosthetic group

(4) Both 1 and 2

64) **Statement I :** There are three major ways in which different cells handle pyruvic acid produced by glycolysis.
Statement II : The reducing agent is NAD^+ which is oxidized to NADH + H^+ in both alcoholic fermentation and lactic acid fermentation.

(1) Both Statement I and Statement II are correct.

(2) Both Statement I and Statement II are incorrect

(3) Statement I is incorrect while Statement II is correct

(4) Statement I is correct while Statement II is incorrect.

65) Read the following four statements A, B, C and D and select the right option having both correct statements.

Statements :

(A) Cyclic photophosphorylation takes place probably to meet the difference in number of ATP and $NADPH.H^+$ used in the dark reaction.

(B) Plastoquinone removes many protons from the stroma while transporting an electron.

(C) Phosphorylation take place only in light reaction but not in dark reaction.

(D) Stroma lamellae have PS-I only and lack NADP reductase.

Options :

(1) C and D

(2) A and B

(3) B and C

(4) A and D

66) Consider the following statements and select the incorrect statement :-

(1) Axillary bud is derived from shoot apical meristem

(2) In sunflower stem, sclerenchymatous pericycle is located just above phloem bundle in semi lunar patches

(3) In conjoint vascular bundle phloem usually located on the inner side of xylem

(4) Phloem parenchyma is usually not present in monocot stem.

67) How many of the following tissue structures are not present in dorsiventral leaf ?
Epidermis, hyopdermis, endodermis, pericycle, pith, medullary rays.

(1) Six

(2) Five

(3) Four

(4) One

68) **Statement-I :** Recessive trait is seen due to non functional enzyme or because no enzyme is produced.

Statement-II : A true breeding line is one that, having undergone continuous self pollination, shows the stable trait inheritance.

(1) Both Statement-I and II are correct.

(2) Both Statement-I and II are incorrect.

(3) Statement-I is correct but II is incorrect.

(4) Statement-I is incorrect but II is correct.

69) A heterozygous round seeded pea plant is crossed with wrinkled seeded pea plant. How many

plants produces intermediate sized starch grain out of 2000 progenies obtained ?

(1) 0
(2) 1500
(3) 1000
(4) 750

70) A husband and wife have normal vision but fathers of them were colour blind. Probability of their first daughter to be colour blind is

(1) 25%
(2) 50%
(3) 15%
(4) 0%

71) Through ___(A)___ and domestication from ancestral wild cows, we have well known Indian breeds, e.g. shahiwal cows in ___(B)___ :

(1) (A) Natural selection ; (B) Punjab
(2) (A) Artificial selection ; (B) Punjab
(3) (A) Natural selection ; (B) Gujrat
(4) (A) Artificial selection ; (B) Gujrat

72) Select the incorrect statement -

(1) Yeast is used for production of ethanol
(2) Enzyme lipases are used in detergent
(3) Bottled juices are made clearer by use of streptokinase
(4) Lipases removes oil stains

73) The ladybird and dragon flies are useful to get rid of __a__ and __b__.

(1) a-Aphids, b-Mosquitoes
(2) a-Mosquitoes, b-Butterfly
(3) a-Butterfly, b-Bacillus
(4) a-Bacillus, b-Mosquitoes

74) Consider following genetic disorders :
Haemophilia, Sickle cell Anaemia, cystic fibrosis, Thalassemia, Phenylketonuria, Polydactyly, Klinefelter's syndrome, Turner's syndrome.
How many of the above disorder are Mendelian disorder ?

(1) 6
(2) 2
(3) 8
(4) 3

75) Fill in the blanks :-
Regulation of lac Operon by_____ is referred to as _____ regulation.

(1) Repressor, Negative
(2) Negative, Repressor
(3) Repressor, Positive
(4) Inducer, Positive

76) After the formation of the product in the bioreactors, it undergoes through separation and purification processes before a finished product is ready for marketing. These processes are collectively referred to as :

(1) Upstream processing
(2) Downstream processing
(3) Elution
(4) Transformation

77) Due to ampicillin resistance gene, one is able to select a transformed cell in the presence of ampicillin. The ampicillin resistance gene in this case is called :

(1) Recombinant gene
(2) Selectable marker
(3) Origin of replication
(4) Recognition site

78) The proteins encoded by the genes cry I Ac and cry II Ab are effective against :-

(1) Corn borer
(2) Cotton bollworms
(3) Nematodes
(4) Mosquitoes

79) Identify the correct statement with respect to development of transgenic animals for study of diseases :-

(1) Transgenic animals are designed to increase our understanding of role of genes in disease.
(2) Animals are developed as model for human diseases to investigate new methods of treatments
(3) Models exist for alzheimer's and cystic fibrosis
(4) All of these

80)

	Column-I		**Column-II**
I.	PCR	A.	Large scale culture
II.	Bioreactor	B.	To induce alien DNA in host cell

III.	Gene gun	C.	Restriction endonuclease
IV.	Eco RI	D.	Amplification of gene

(1) I - D, II - A, III - B, IV- C

(2) I - B, II - A, III - D, IV - C

(3) I - D, II - A, III - C, IV - B

(4) I - A, II - D, III - B, IV - C

81) How many different genotype and phenotype are possible in the ABO blood group typing in human beings respectively :-

(1) 6 & 4

(2) 6 & 6

(3) 4 & 6

(4) 4 & 4

82) Match the following column and select the correct option :-

(a)	Lady bird	(i)	Aphids
(b)	*Bacillus thuringiensis*	(ii)	Butter flies caterpillars
(c)	Baculovirus	(iii)	Insect and other arthropods
(d)	*Trichoderma*	(iv)	Effective against several plant pathogen

(1) a-i, b-ii, c-iii, d-iv

(2) a-i, b-iii, c-iv, d-ii

(3) a-i, b-iv, c-iii, d-ii

(4) a-ii, b-iv, c-i, d-iii

83) Match the term given in column I with their appropriate meaning in column II and select the most suitable option :-

(a)	Allele	(i)	F_1 always resemble either of the parents
(b)	Co dominance	(ii)	F_1 was in between the parents
(c)	Incomplete dominance	(iii)	F_1 resemble both parents
(d)	Dominance	(iv)	Slight different form of the gene

(1) a-iv, b-iii, c-ii, d-i

(2) a-i, b-ii, c-iii, d-iv

(3) a-ii, b-iv, c-i, d-iii

(4) a-iii, b-i, c-iv, d-ii

84) Unsegmented body with distinct head, muscular foot and visceral hump is seen in :-

(1) *Pila*

(2) *Ascaris*

(3) *Cucumaria*

(4) Both (1) and (2)

85) Read the following statements and choose the correct set of statements :
(a) RNA was the first genetic material
(b) RNA polymerase binds to start codon and initiates translation.
(c) The cellular factory responsible for synthesising proteins is the ribosome
(d) In lac operon polycistronic structural gene is regulated by a common promoter and regulatory genes
Choose the correct answer from the options given below :

(1) (a) and (c) only

(2) (a), (b), (c) and (d) all

(3) (a), (c) and (d) only

(4) (a) and (d) only

86) Match List-I with List-II :

	List-I		**List-II**
(a)	Release factor	(i)	Splicing
(b)	SnRNA	(ii)	Translation
(c)	Probe	(iii)	Transcription
(d)	Rho factor	(iv)	DNA fingerprinting

(1) (a)-(iii), (b)-(ii), (c)-(iv), (d)-(i)

(2) (a)-(iii), (b)-(iv), (c)-(ii), (d)-(i)

(3) (a)-(ii), (b)-(i), (c)-(iv), (d)-(iii)

(4) (a)-(iv), (b)-(ii), (c)-(i), (d)-(iii)

87) The z gene codes for beta-galactosidase (β-gal), which is primarily responsible for the hydrolysis of the disaccharide into :

(1) glucose and glycogen

(2) galactose and glucose

(3) glucose and fructose

(4) glucose and lactose

88) Which of the following is a not a correct statement with respect to polymorphism :

(1) Polymorphism arises due to mutations

(2) Polymorphisms ranging from single nucleotide change to very large scale changes

(3) For evolution and speciation, polymorphisms play very important role

(4) The probability of polymorphism to be observed in coding DNA sequence would be higher

89) In dsDNA adenine pair with thymine and guanine pair with cytosine by how many hydrogen bonds respectively ?

(1) 1, 2

(2) 2, 3

(3) 2, 1

(4) 3, 2

90) A plant having the genotype AABbCC will produce ____ kinds of gametes :

(1) 5

(2) 4

(3) 3

(4) 2

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	2	2	4	3	1	1	3	2	3	4	4	3	4	3	2	1	2	3	2	2
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	2	4	2	4	1	2	4	4	1	1	3	2	1	4	3	1	3	3	2	1
Q.	41	42	43	44	45															
A.	3	3	2	4	2															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	3	3	4	4	3	1	1	2	2	1	2	2	4	1	1	1	4	4	4	4
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	3	3	4	3	3	1	1	4	2	2	2	2	1	4	1	2	1	3	2	2
Q.	86	87	88	89	90															
A.	1	4	3	2	3															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	2	3	3	3	1	4	3	2	3	3	3	1	3	1	4	2	2	1	2	3
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	1	1	3	4	1	1	4	2	3	3	3	1	1	2	3	3	3	1	3	2
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	1	3	4	4	2	3	3	4	4	2	1	3	3	1	4	1	3	1	2	3
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	4	4	3	4	4	3	2	1	3	4	2	3	1	1	1	2	2	2	4	1
Q.	171	172	173	174	175	176	177	178	179	180										
A.	1	1	1	1	3	3	2	4	2	4										

SOLUTIONS

PHYSICS

1) Here, $u = 144 \text{ km h}^{-1} = 144 \times \frac{5}{18} \text{ ms}^{-1} = 40 \text{ ms}^{-1}$
$v = 0, \ S = 200 \text{ m}$
As $v^2 - u^2 = 2aS$
$(0)^2 - (40)^2 = 2 \times a \times (200)$
$a = -\frac{(40)^2}{2 \times 200} = -4 \text{ ms}^{-2}$
As $v = u + at$
$0 = 40 - (4)(t)$
$t = \frac{40}{4} = 10\text{s}$
OR

$$S = \left(\frac{u+V}{2}\right) t$$

$$200 = \left[\frac{40+0}{20}\right] t$$

$$t = \frac{200 \times 2}{40}$$

$t = 10$ sec

2) For triangular lamina $I = \frac{m\ell^2}{12}$

3)

As the inclined plane is smooth, the sphere can never roll rather it will just slip down.

Hence, the angular momentum remains conserved about any point on a line parallel to the inclined plane and passing through the centre of the ball.

4) As $T = m\omega^2 R$
$\Rightarrow \quad 400 = 1 \times t^2 \times 1$
$\Rightarrow \quad t = 20$ sec.

5)

$g_{eff} = g - \omega^2 R$ on equator
If $g_{eff} = 0 \Rightarrow \omega^2 = \frac{g}{R}$

$$\omega^2 = \frac{10}{6400 \times 10^3} = \frac{1}{64 \times 10^4} = \left(\frac{1}{800}\right)^2$$

$\omega = \frac{1}{800}$ rad/s

6)

As per theory
for temperature change
Q = msΔT
for phase change
Q = mL

7) $\frac{E_{star}}{E_{sun}} = \left(\frac{A_{star}}{A_{sun}}\right)\left(\frac{T_{star}}{T_{sun}}\right)^4$

$10000 = \left(\frac{r^2_{star}}{r^2_{sun}}\right)\left(\frac{1}{3}\right)^4$

$\Rightarrow \frac{r_{star}}{r_{sun}} = \frac{900}{1}$

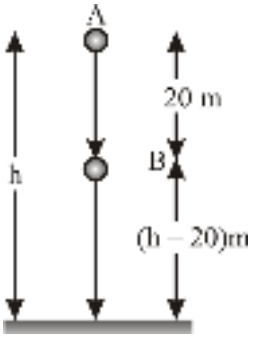
8) $\eta = \left(1 - \frac{T_2}{T_1}\right) \times 100$

$\Rightarrow \frac{T_2}{T_1} = 0$

(T_2 = 0K)

9) For first ball h = $\frac{1}{2}gt^2$ ...(i)

For second ball $(h-20) = \frac{1}{2}g(t-1)^2$ ...(ii)



subtract equation (ii) from (i)
20 = –5 + 10t ⇒ t = 2.5 sec
Hence, height $h = \frac{1}{2} \times 10 \times (2.5)^2 = 31.2m$

10)

Slope at C is maximum

11) $U_i = -\frac{M}{3}g\frac{L}{6} = -\frac{MgL}{18}$ $U_f = 0$

$W_{ext} = -|\Delta U| = U_f - U_i = \frac{MgL}{18}$

12) Force = Rate of change of momentum

Initial momentum $p_1 = mv \sin\theta \hat{i} + mv \cos\theta \hat{j}$

Final momentum $p_2 = -mv \sin\theta \hat{i} + mv \cos\theta \hat{j}$

$$F = \frac{\Delta p}{\Delta t}$$

$$= \frac{-2mv\sin\theta}{2 \times 10^{-3}}$$

Substituting m = 0.1 kg, v = 5 m/s, θ = 60°

Force on the ball $F = -250\sqrt{3}$N

Negative sign indicates direction of the force.

13)

Liquid with low density float over high density.

14) $W = 8\pi T(r_2^2 - r_1^2) = 8\pi T\left[\left(\frac{2}{\sqrt{\pi}}\right)^2 - \left(\frac{1}{\sqrt{\pi}}\right)^2\right]$

$W = 8 \times \pi \times 30 \times \frac{3}{\pi} = 720$ erg

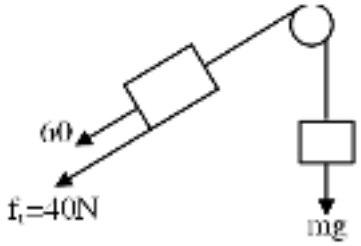
15) $T = 2\pi\sqrt{\frac{m}{K}} \Rightarrow K \propto \frac{1}{T^2}$

In this case $K = K_1 + K_2$

$\frac{1}{t_0^2} = \frac{1}{t_1^2} + \frac{1}{t_2^2} \Rightarrow t^{0^{-2}} = t_1^{-2} + t_2^{-2}$

16)

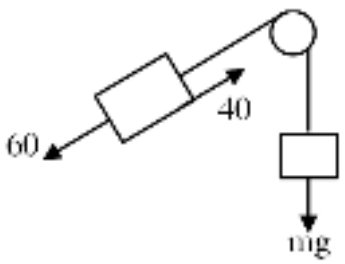
For maximum 'm'



mg = 100

m = 10 kg

For minimum 'm'



mg + 40 = 60

m = 2kg

17) $y_1 = a \sin(kx - \omega t)$
$y_2 = a \sin(kx + \omega t)$
According to the principle of superposition, the resultant wave is
$y = y_1 + y_2$
$= a \sin(kx - \omega t) + a \sin(kx + \omega t)$
Using trigonometric identity
$\sin(A + B) + \sin(A - B) = 2 \sin A \cos B$
we get
$y = 2a \sin kx \cos \omega t$

18) $\frac{a_1}{a_2} = \frac{4}{3} \Rightarrow \frac{I_1}{I_2} = \frac{a_1^2}{a_2^2} = \frac{16}{9}$

$$\frac{I_{max}}{I_{min}} = \frac{\left(\sqrt{I_1} + \sqrt{I_2}\right)^2}{\left(\sqrt{I_1} - \sqrt{I_2}\right)^2} = \frac{49}{1}$$

19) $$m_{mean} = \frac{2.46 + 2.47 + 2.41 + 2.42 + 2.44}{5}$$
$m_{mean} = 2.44$ kg
$\Delta m_1 = 2.44 - 2.46 = -0.02$ kg
$\Delta m_2 = -0.03$ kg, $\Delta m_3 = 0.03$ kg, $\Delta m_4 = 0.02$ kg
$\Delta m_5 = 0.00$ kg
Mean error
$= \frac{1}{5} [|-0.02| + |-0.03| + 0.03 + 0.02 + 0.00]$
$= 0.02$ kg m = 2.44 ± 0.02 kg

The weight of straw will be balanced by the force of surface tension $\therefore mg = 2Tl \Rightarrow$

$$m = \frac{2Tl}{g}$$

20) $\frac{2 \times 3 \times 10^{-2} \times 10 \times 10^{-2}}{9.8}$ kg $0.6gm$

$$a = \frac{100}{10 + 10 + 5} = 4 m/s^2$$

21) force on B = 5 × a = 5 × 4 = 20 N

$$\lambda_m \propto \frac{1}{T} \Rightarrow \lambda m_1 T_1 = \lambda m_2 T_2$$

23) $\Rightarrow T_2 = \frac{\lambda m_1 T_1}{\lambda m_2} = \frac{1.4 \times 10^{-6} \times 1000}{2.8 \times 10^{-3}} = 500K$

24) Percentage error in A

$$= \left[\begin{matrix} 2\times 0.1 + 3\times 0.3 \\ +1\times 0.2 \\ +\frac{1}{2}\times 0.2 \end{matrix}\right]\%$$

= 1.4%

25)

for dia magnetic

$\mu_r < 1$, $\chi > 0$

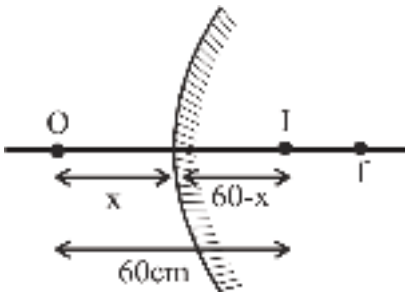
26)

$$R = \frac{\sqrt{2mk}}{qB}$$

$$R' = \frac{\sqrt{2m(2k)}}{q(3B)}$$

$$\frac{R'}{R} = \frac{\sqrt{2}}{3}$$

27)



$$m = -\frac{v}{u}$$

$$\frac{1}{4} = -\frac{60-x}{-x} \quad \Rightarrow x = 48cm$$

u = 48 cm

v = 12 cm

$$\frac{1}{u} + \frac{1}{v} = \frac{1}{f}$$

$$\frac{1}{-48} + \frac{1}{12} = \frac{1}{f}$$

f = 16 cm

28) ${}_{90}X^{200} \xrightarrow{n\alpha,\, n'\beta} {}_{80}Y^{168}$

$$n = \frac{\Delta A}{4} = \frac{200-168}{4} = \frac{32}{4} = 8$$

${}_{90}X^{200} \xrightarrow{8\alpha} {}_{74}Z^{168} \rightarrow {}_{80}Y^{168}$

n' = 80 – 74 = 6

29)

microwave oven is based on microwave.

30) $a \sin\theta = 2\lambda$

$$\lambda = \frac{a\sin\theta}{2} = \frac{24 \times 10^{-7} \times \sin 30^\circ}{2}$$

$= 6000$ Å

31) $$i = \frac{V}{R} = \frac{10}{2} = 5A$$

$$U = \frac{1}{2} Li^2 = \frac{1}{2} \times 2 \times 25 = 25 \text{ J}$$

32) $$f = \frac{1}{2\pi}\sqrt{\frac{1}{80 \times 10^{-6} \times 5}} = \frac{25}{\pi}$$

33) $\theta = (\mu_v - \mu_R) A = (1.68 - 1.56) \times 12^\circ = 1.44^\circ$

34) In Ist case : $H = I^2Rt \Rightarrow H \propto I^2t$; $t_1 = 20$ seconds, $t_2 = 18$ seconds

$$\Rightarrow \quad \frac{H_2}{H_1} = \left(\frac{I_2}{I_1}\right)^2 \frac{t_2}{t_1}$$

$$\Rightarrow \quad \frac{H_2}{600} = \left(\frac{4}{3}\right)^2 \times \frac{18}{20} \Rightarrow H_2 = 960J$$

35) As inward B decreases, so CW current in bigger loop & ACW current in smaller loop.

36) $$\int_0^A dV = -\int_0^2 E dx$$

$$V_A - V_0 = -\int_0^2 30x^2 dx = -30\left[\frac{x^3}{3}\right]_0^2$$

$$= -10 \times \frac{8}{3}$$

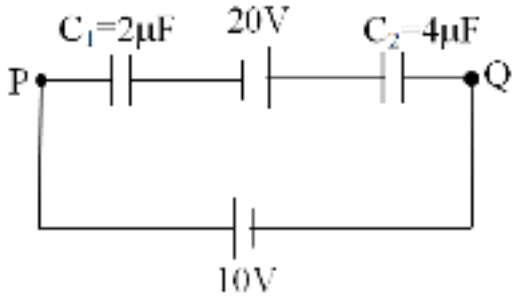
$= -80$ Volt

37)

$I = neAv = ne\pi r^2 v$

$$I' = ne\pi\left(\frac{r}{2}\right)^2 (2v) = \frac{I}{2}$$

38) Both battery are in series & both capacitors are also in series as per shown figure.



$\square$ $V_{C_1} = \frac{Q}{C_1} = \frac{C_{eff} \times V_{eff}}{C_1}$

$= \frac{\frac{C_1C_2}{C_1+C_2} \times Veff}{C_1} = \frac{C_2}{C_1 + C_2} \times V_{eff}$

$= \frac{4}{2+4} \times 30 = 20$ V

39) $K_{max}(eV) = 12400 \left[\frac{1}{\lambda(Å)} - \frac{1}{\lambda_0(Å)}\right]$ ev.Å

$= 12400\left[\frac{1}{1000} - \frac{1}{2000}\right] = 6.2$ eV

40) $\lambda = \frac{h}{\sqrt{3mKT}}$

$\lambda \propto \frac{1}{T}$

$\frac{\lambda_2}{\lambda_1} = \sqrt{\frac{T_1}{T_2}}$

$\lambda_2 = \lambda\sqrt{\frac{300}{1200}}$

$= \frac{\lambda}{2}$

41) $n\lambda_1 = (n + 1)\lambda_2$

$7500\ n = 6000\ (n + 1)$

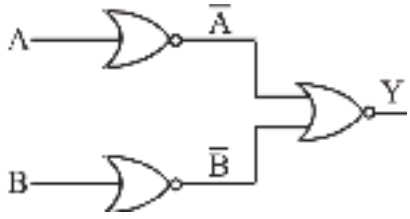
$5n = 4n + 4$

$n = 4$

42)

$R \propto A^{1/3} \Rightarrow \frac{R_1}{R_2} = \left(\frac{16}{128}\right)^{1/3} \Rightarrow \frac{3 \times 10^{-15}}{R_2} = \frac{1}{2}$

$R_2 = 6 \times 10^{-15}$ m

43) 

$Y = \overline{\bar{A} + \bar{B}}$

$Y = \bar{A} + \bar{B} = A.B$ (AND)

44) $\sin\theta_C = \frac{V_D}{V_r} = \frac{1.8}{2.4} = \frac{3}{4}$

$\theta_C = \sin^{-1}\frac{3}{4}$

45) $I_{Load} = \frac{6}{1 \times 10^3} = 6mA$

$I_{100\Omega} = \frac{9-6}{100} = 30$ mA

$I_Z = I_{100} - I_L = 30 - 6 = 24$ mA

CHEMISTRY

46) $\alpha = \sqrt{\frac{K_a}{C}}$

$\Rightarrow \alpha = \sqrt{\frac{1.4 \times 10^{-5}}{(0.1/2)}}$

$\Rightarrow \alpha = 1.67 \times 10^{-2}$

47)

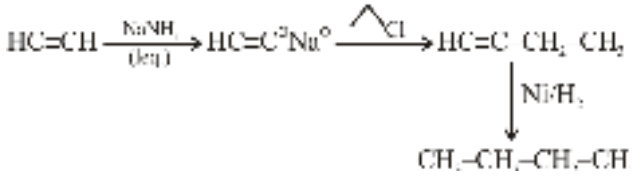
48) Charge required to deposite 1 gm equiv. of any substance is 1F

49) V.P. $\propto \frac{1}{b.p.}$

Solution III has lowest

Value of V.P. so highest b.p.

50) Geometrical isomers are diastereomers also

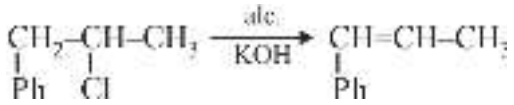
51) 

52)

As per trans effect.

53)

54)

55) 

56)

B.E. ∝ Stability

57)

Stability in aqueous
$Cr^{3+} > Mn^{3+}$
$CrO_4^{2-} \xrightarrow{H^+} Cr_2O_7^{2-}$
yellow orange
Brass = Cu + Zn
Bronze = Cu + Sn

58)

Concept

59)

Question Explanation: Identify which given compounds/species are *chemically possible (can exist)*.

Concept: Octet rule, expansion of octet, stability of ions.

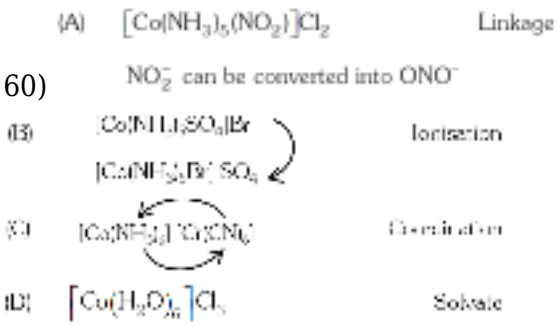
Solution:

A. **H_2S** → Exists (bent molecule).
B. **OF_6** → Does *not* exist (O cannot expand octet beyond 8e⁻).
C. **BF_4^-** → Exists (stable tetrahedral ion).
D. **PH_5** → Does *not* exist (P–H bonds cannot stabilize hypervalent form).
E. **XeF_5** → Does *not* exist (XeF_4 and XeF_6 exist, not XeF_5).
F. **CCl_6^{2-}** → Does *not* exist (carbon can't form six bonds).
G. **PCl_5** → Exists (stable in gaseous state, trigonal bipyramidal).

Existing species: H_2S, BF_4^-, PCl_5 → **3 compounds**

Final Answer: Option (1)

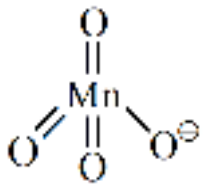
60)



61)

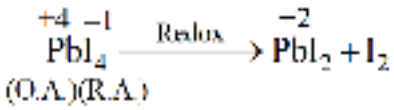
$KMnO_4$

$K^+ MnO_4^-$



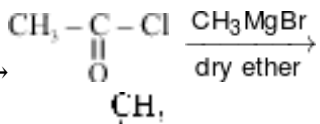
Mn^{+7} → d° configuration
diamagnetic

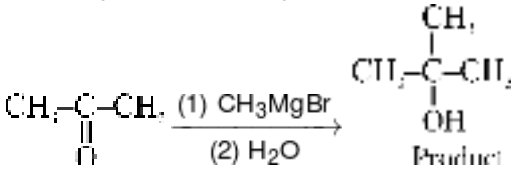
62)



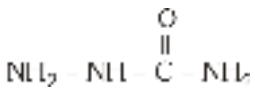
63)

64) $CH_3COOH + PCl_5 \rightarrow$

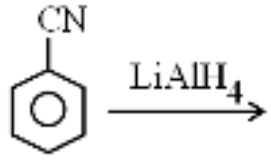




65)

Reagent : 

Semicarbazide

66) 

67)

d > c > a > b

68) NCERT 11th P.No. 374

69)

Fact.

70)

(i) & (iii) only

71)

$K = \sqrt{\frac{K_1}{K_2}}$

72) **Generated by Allie**

Problem Statement: The question asks to match terms related to collision theory in chemistry. It requires linking definitions or descriptions in Column-I with the correct terms or concepts in Column-II that describe aspects of molecular collisions and reaction kinetics.

Underlying Concept: The core concept is collision theory in chemical kinetics. It explains how chemical reactions occur through collisions between molecules. Key ideas include collision frequency, activation energy, effective collisions, and deviations for complex molecules.

Relevant Formulas: The relevant formula is the rate expression from collision theory:
\[\text{Rate} = Z_{AB} e^{-E_a / RT} \]

Where: - Z_{AB} = collision frequency (number of collisions per second per unit volume) - E_a = activation energy - R = gas constant - T = temperature - $e^{-E_a/RT}$ = fraction of molecules with energy equal to or greater than E_a (Boltzmann factor)
Effective collisions lead to reaction, depending on sufficient kinetic energy and proper orientation.

Tips and Tricks: Remember the order: number of collisions (frequency), fraction energetic enough (Boltzmann factor), adjustments for complex molecules, then effective collisions that cause reactions.

Common Mistakes: Confusing collision frequency with effective collisions. Not identifying the Boltzmann factor as the fraction of molecules with enough energy. Overlooking deviations caused by complex molecules.

Why Other Options Are Incorrect?: Options mixing collision frequency with effective collisions or mismatching complex molecules lead to misunderstanding of reaction kinetics and collision theory.

73) **Explain question:**
To select the correct match.
Concept: Azeotropic Mixture

Solution:
(A) 68% HNO_3+ 32% H_2O → This mixture forms a maximum boiling azeotrope because it shows negative deviation from Raoult's law. (A → Ⅲ).
(B) Solution having large positive deviation: it form minimum boiling azeotropes (B → I).
(C) n-heptane and n-hexane : It forms ideal solution (C → Ⅱ).
(D) Two solutions having the same osmotic pressure at a given temperature: are known as isotonic solution (D → IV).

Final answer: Option (4)

74) $\Lambda_m^o(NH_4OH) = \Lambda_m^o(NH_4Cl) + \Lambda_m^o(NaOH) - \Lambda_m^o(NaCl)$
$\alpha = \frac{\Lambda_m}{\Lambda_m^o}$

75)

76)

77)

78)

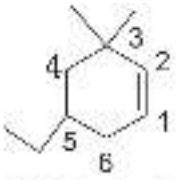
79)

$Mn^{+2} \rightarrow 3d^5 \rightarrow$ stable

80) $Mn^{+2} \rightarrow d^5$

1	1	1	1	1

81) Colour of $BaCrO_4$ is yellow
H_2S is a weak acid it does not produce free S^{2-} ion



82) 5-ethyl-3,3-dimethyl cyclohexane

83)

84)

C_2H_5–NH–C_2H_5 is a secondary amine, therefore, it will not give carbylamine test.

85) $\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$

$V_2 = \frac{P_1V_1T_2}{P_2T_1} = \frac{700 \times 40 \times 273}{760 \times 300}$

$V_2 = 33.52$

22400mL of N_2 consis 28 g then 33.52 mL g N_2 confain

$\frac{33.52 \times 28}{22400}$

$N\% = \frac{0.0419}{0.25} \times 100$

=16.76

86)

87)

88) For zero order reaction
$[A]_t = [A]_0 - Kt$

89) $\frac{P_A^o - P_S}{P_S} = \frac{n_B}{n_A}$
$\Delta T_f = K_f.m$

90)

BIOLOGY

91) NCERT Pg. # 102

92)

93) NCERT Page No. 117

94)

95) NCERT (XI) Pg. # 51

96) NCERT XI, Page # 312

97)

98)

99)

100)

101) NCERT XI - Pg. No. 318

102) NCERT-XI, Pg # 317

103)

104)

105)

106) NCERT Page No. # 225 (E/H)

107)

108) NCERT Pg # (E) 332, (H) 333

109) NCERT Page # 51

110) NCERT XII, Page # 127

111) NCERT XII, Page # 130

112) NCERT Pg # (E) 332, (H) 333

113) NCERT-XII, Pg # 48

114) NCERT-XII, Pg # 48

115)

NCERT (XIIth) (Eng), Para-2, Pg. # 54

NCERT (XIIth) (Hindi), Para-1, Pg. # 59

116)

NCERT XII, Pg. # 152 (E), 166 (H)

117)

118)

NCERT-XII Pg. # 157

119) NCERT-XII, Pg # 53

120) NCERT Page No. # 150

121) NCERT Pg. # 245

122) NCERT Pg. # 242

123) NCERT XII Pg # 237

124)

125)

126) NCERT Pg.# 233, 236, 251 [E]
253, 257, 273 [H]

127) NCERT-XI, Pg. # 126

128) **Question Explanation :**
Question asking about the ribosomes

Concept :
Question is based on ribosomes

Solution :
The eukaryotic ribosomes are 80S Each ribosome has two subunits, larger and smaller subunits .The two subunits of 80S ribosomes are 60S and 40S.

Final Answer :
Option (1)

129)

Explaining the Question :- The question asks about the specific phase in the cell cycle during which **DNA replication** and **centriole duplication** take place.
Concept:- This question is based on **Cell cycle**
Solution :- DNA replication and centriole duplication occur during the **S phase** of the cell cycle.
Final Answer :- (3)

130) NCERT XI Pg # 140

131) NCERT Pg # 167, 170

132) NCERT XII Pg # 243

133) NCERT XII Pg # 262

134) NCERT Page No. # 126, 127

135) NCERT XI Pg # 78, 80-81

136) NCERT XI Pg # 73

137)

Explain Question : The question asks you to identify which of the listed processes is not a part of sexual reproduction.
Concept : This question is based on gymnosperm, general characters and life cycle.
Explanation : Formation of new organism from unfertilized female gamete - Not sexual reproduction
This describes parthenogenesis.
Final answer : 3

138) NCERT XII Pg # 23, 25, 36, 38

139) NCERT XI Pg # 220

140)

Question Explanation: Question asks about the kingdom whose organism boundaries are not well defined.

Concept: Protista

Solution: Kingdom Protista contains unicellular eukaryotic organisms. Their boundaries are unclear because they show mixed characteristics.

Final Answer: Option (2) Protista

141) NCERT XI Pg # 38

142) NCERT XI Pg # 33, 35, 36, 38

143) NCERT XII Pg # 34, 35, 36, 38
Hint : Statement A, B, D and E are correct.

144) NCERT XI Pg # 222 Fig. 13.10

145) NCERT Pg # 232, fig-14.3

146) NCERT Pg # 248

147) NCERT XI Pg # 23, 24

148) NCERT XI Pg # 34, 35, 36, 44

149) NCERT, Page # 146

150) NCERT XI, Page # 144

151) NCERT XI, Page # 145

152) NCERT XI Page No. # 247.

153) NCERT-XI, Pg. # 159

154)

NCERT-XI, Pg. # 230

155) NCERT-XI, Pg # 213,214,217,218

156) NCERT-XI, Pg. # 85, 91, 90 only c is wrong

157) NCERT Page : 93
Hypodermis, endodermis, pericycle, pith and medullary rays.

158) NCERT Pg. # 70, 77

159) NCERT Page-71

160) NCERT Page-289

161) NCERT-Pg : 70

162) NCERT Page - 183

163)

164) NCERT Page-88

165) NCERT (XII) Pg. # 117

166)

167)

168)

169)

170)

171) NCERT XII Pg # 77

172) NCERT XII Pg # 187

173) NCERT XII Pg # 76, 77

174) NCERT, Page # 53 (E), 53 and 54 (H)

175)

NCERT XII Page No. # 104, 115, 116

176)

NCERT XII Page No. # 115, 122, 110

177)

NCERT XII Page No. # 116

178)

NCERT XII Page No. # 121, 122

179) NCERT XII Page No. # 97

180)

NCERT-XII, Pg. # 72, 73